

UNIT-2

ENTITY RELATIONSHIP MODEL

1. Explain the basic building blocks of ER model.

ER Model:

- It was introduced by Peterchen in 1976.
- It describes the structure of Database with the help of diagrams, which is known as ERD (Entity Relationship Diagrams).
- It is a blue-print or design of Database that can later be implemented as Database.

Advantages:

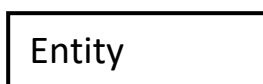
- It is an effective communication tool.
- ER diagrams are very easy to translate into relational tables which can be used to quickly build databases.

Basic building blocks of ER Model:

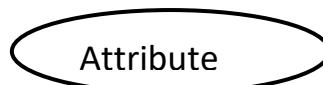
The basic building blocks of ER-Model are

- i. Entity ii. Attributes iii. constraints iv. Relationships

I. Entity: An entity is defined as real world object. It can be anything like a student, faculty, doctor, patient, product etc., An entity stores collection of records. An entity is represented by “rectangle” symbol in a database design.

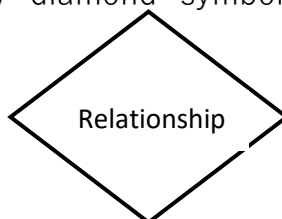


ii.Attribute: Attributes is defined as a character or a group of characters that has a specific meaning. It is also known as “Field”. An attribute is represented by “oval” symbol in a database design.



iii.Constraints: Constraints are “set of rules” which we enforce on data in a database.

iv.Relationship: A relationship is defined as an association among entities. A relationship is represented by “diamond” symbol in a database design.



Types of Relationships: There are three types of relationships they are:

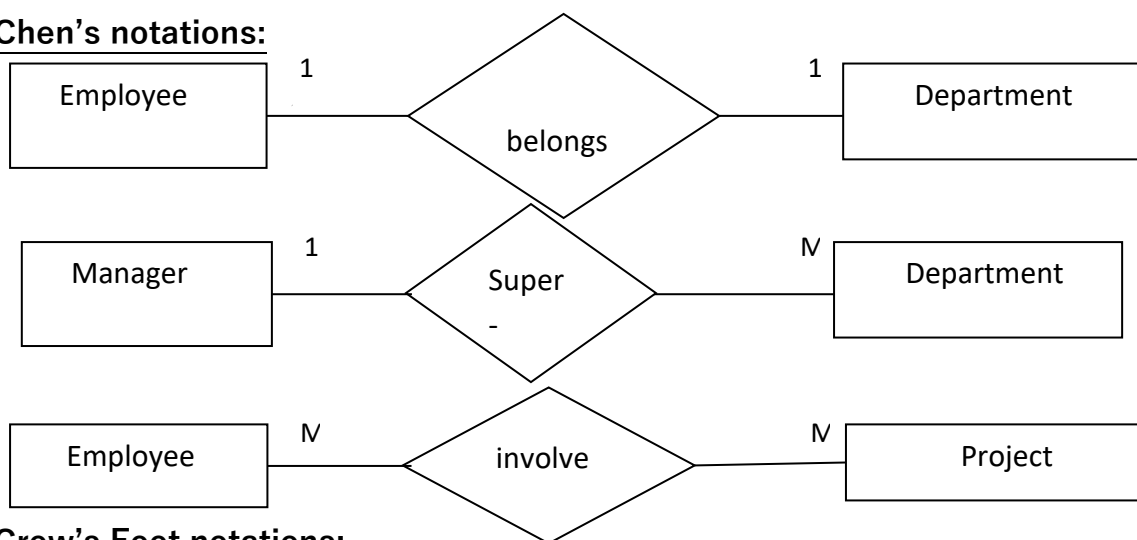
1. one-to-one (1:1 or 1..1)
2. one-to-many (1:M or 1..*)
3. Many-to-one (M:1)
4. many-to-many (M: N or *.*.)

ER Notations: The entities and their relationships are represented with the notations of ER model.

They are two different types of ER notations

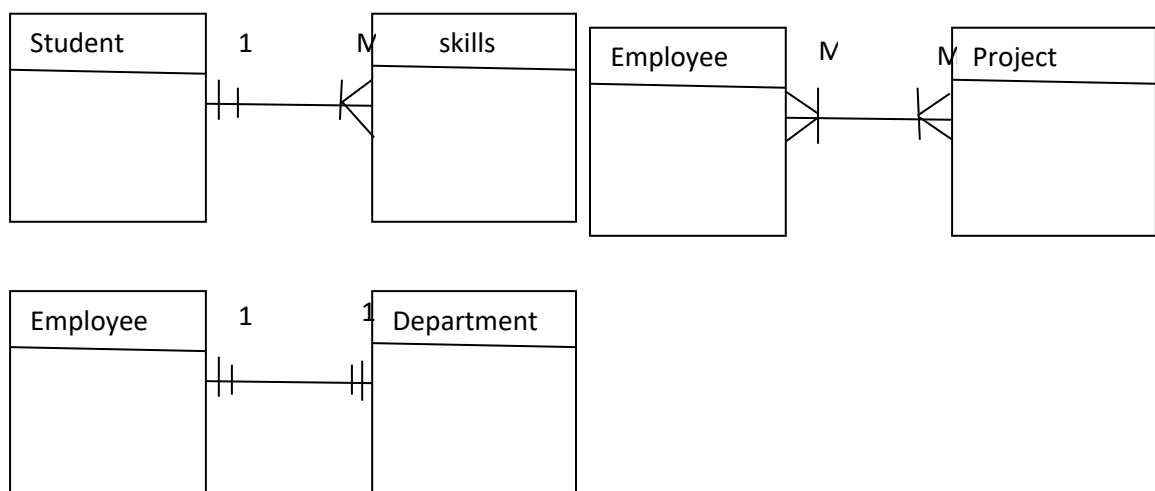
1. Chen's notations
2. Crow's Foot notations

1.Chen's notations:



2.Crow's Foot notations:

- It was developed by Gordan Everest.
- In this notation, the entity is represented with rectangle box.



2.Explain about the classification of Entity sets

Classification of Entity Sets:

Entity: An Entity is an object that is represented in the database.

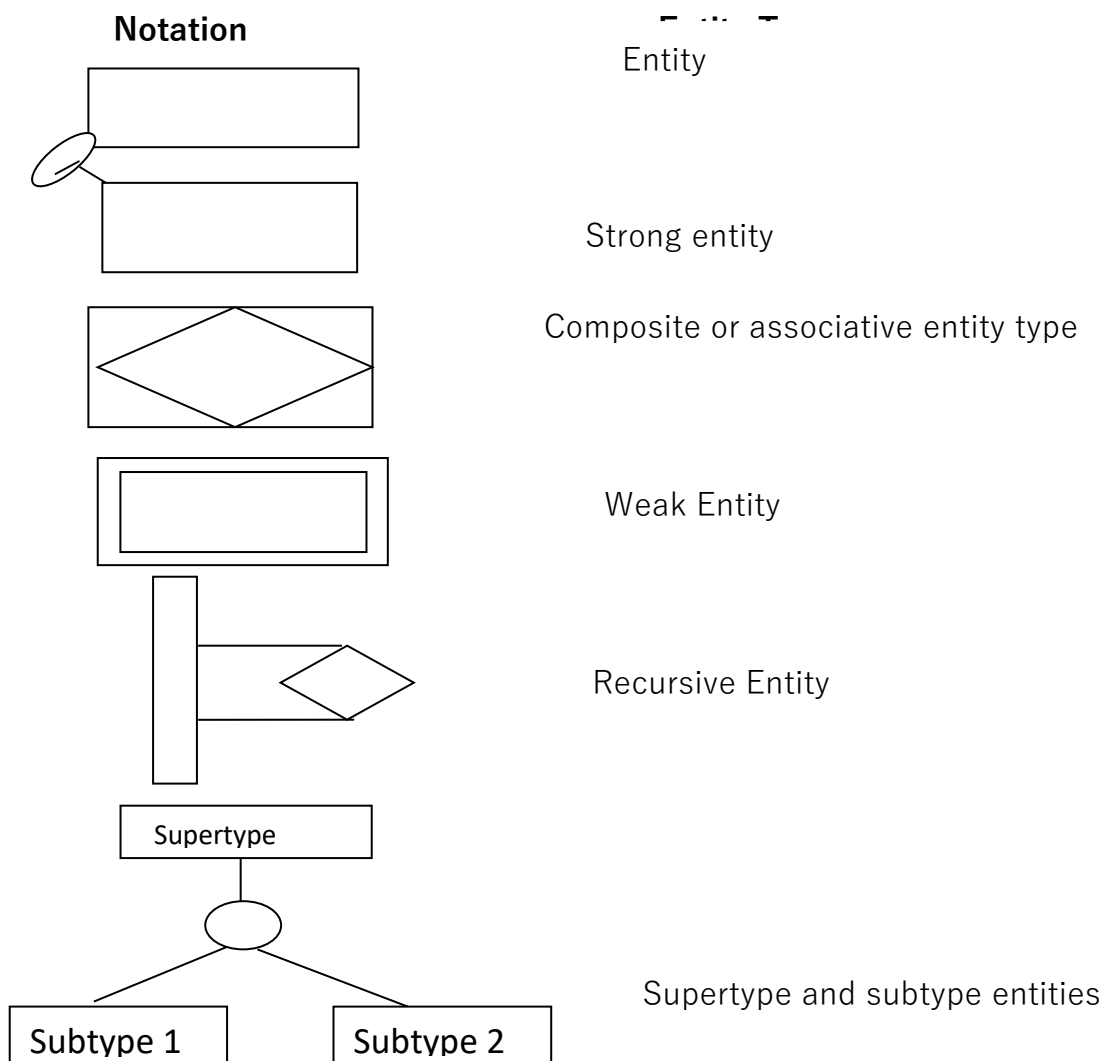
Ex: person, place, thing etc.

Types of Entities:

Entities based on their characteristics are classified as follows:

1. Strong entity type
2. Weak entity type
3. Recursive entity type
4. Composite or Associative entity type
5. Supertype and subtype entities

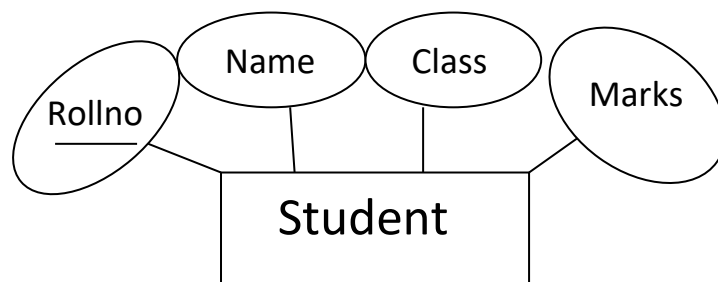
Notations of different entity types in ER diagrams:



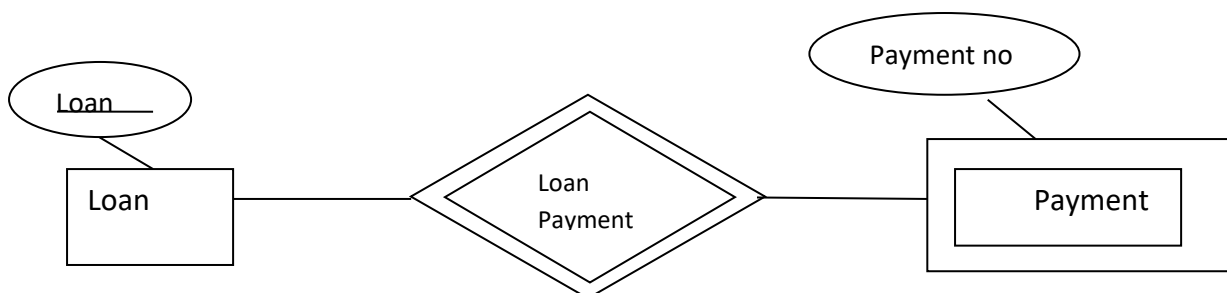
1.Strong Entity type:

- An entity set that has a primary key is called Strong entity.
- It is also called as Regular entity type.
- It is represented by a rectangle.
- It contains a primary key attribute represented with an underline.
- The member of strong entity set is known as Dominant entity set.

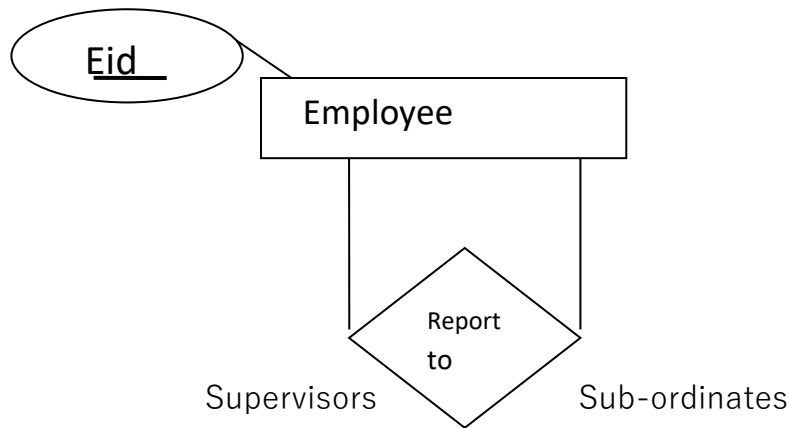
Here, student is an entity and rollno is a primary key attribute. So, it forms a strong entity set.

**2. Weak entity type:**

- An entity set that does not have a primary key is referred as a weak entity set.
- It is represented by double rectangle.
- It contains a partial key represented by a dashed underline.
- The member of weak entity set is called as Sub-ordinate entity set.
- The relationship between one strong and a weak entity set is represented by double diamond sign.

**3. Recursive entity type:**

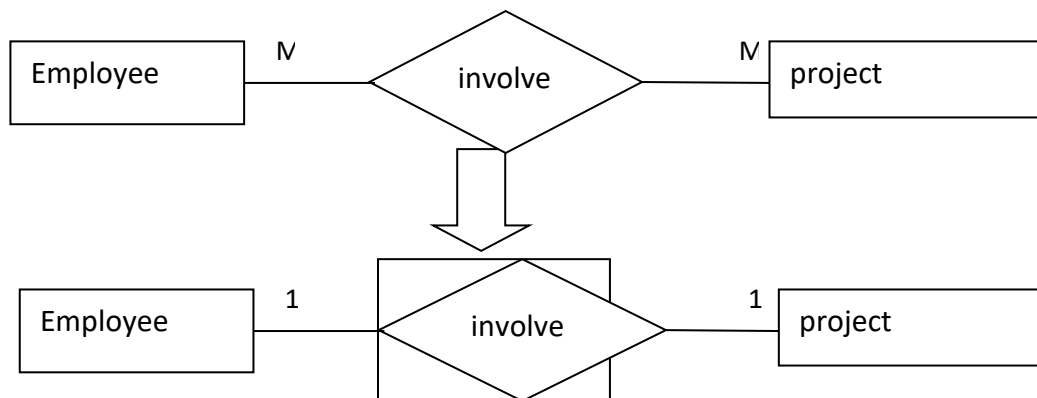
- It is also called as self-referential relationship entity type.
- It is an entity type with foreign key referencing to same table or itself.
- It occurs in a unary relationship



4. Composite entity type:

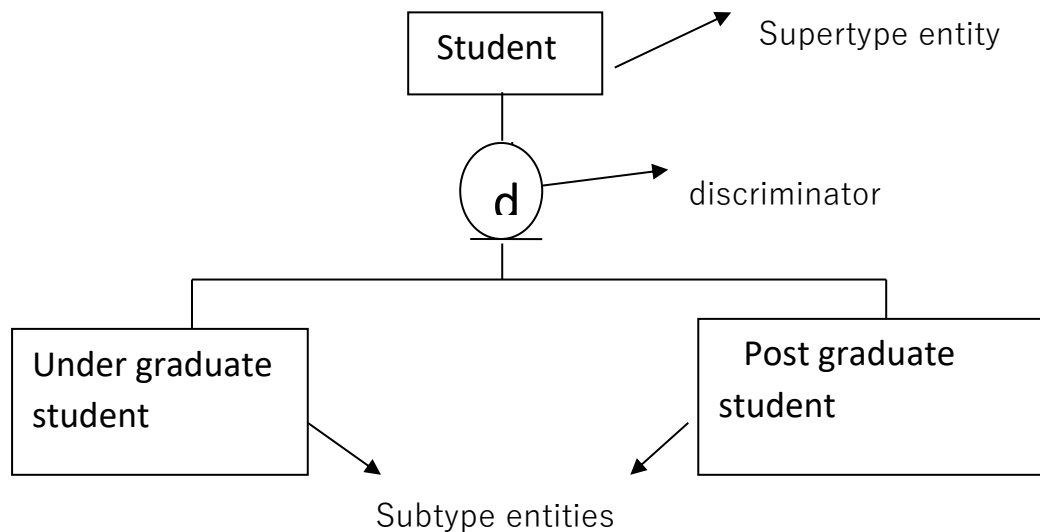
- If many-to-many relationships exist in an ER diagram, then we must eliminate it by using composite entities to reduce the relationship complexity.
- It converts the many-to-many relationship into one-to-many or one-to-one relationship.

Ex: Here, many-to-many causes problems to ERD in converting into relational models. So, it must be converted into other relation as follows:



5. Supertype and Subtype entities:

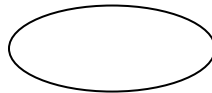
A subtype is a sub-grouping of entities and supertype is a generic entity type that has a relationship with one or more subtypes.



2.Explain about the classification of attributes?

Attributes: -

- It is the name of column.
- It is also called as field (or) data item.
- It represents with oval.



Ex: - A customer of bank may be described by acc number, name, address etc.

- Here, customer is an entity.
- Acc number, name and address are attributes.

Types of Attributes: -

The different types of attributes are:

- Single Valued Attribute
- Multi-Valued Attribute
- Simple / Atomic Attribute
- Compound / Composite Attribute
- Stored Attributes
- Derived Attributes

- Complex Attributes
- Key Attributes
- Non-key Attributes
- Required Attributes
- Optional/Null Value Attributes

1. Single valued Attributes: -

- An attribute that has a single value for a particular entity is known as single valued attribute.

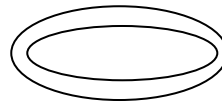
Ex: - Age of an employee entity.

2. Multi-Valued Attributes: -

- An attribute that can have multiple values for the same entity is known as multi-valued attribute.

Ex: - Phone number can have multiple number of the same entity.

- It is represented with double oval.



3. Simple Attributes: -

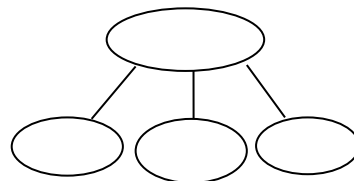
- The attributes which cannot be divided into smaller sub parts are called as simple (or) atomic attributes.

Ex: - The attributes like age, marital status cannot be sub-divided.

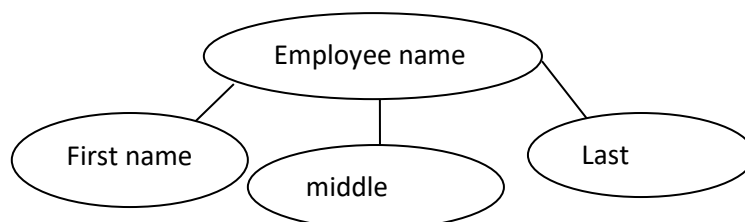
4. Compound Attributes: -

- The attributes which can be divided into smaller sub parts are called as compound or composite attributes.

It is represented with



Ex: - An employee name can be classified into first name, middle name, last name.



5.Stored Attributes: -

- An attribute that cannot be derived from other attributes is called as stored attribute.

Ex: - Date of birth of an employee is a stored attribute.

6.Derived Attributes: -

- The attributes that are derived from other attributes are called as derived attributes.

Ex: - Age can be derived by the difference between current date and date of birth.

- It is represented with-

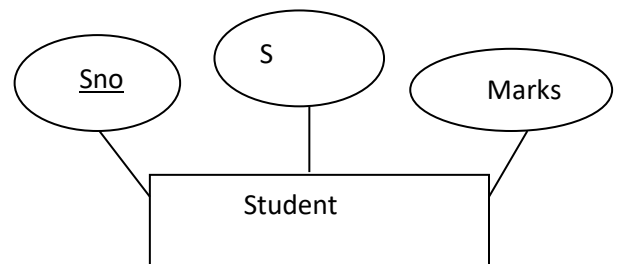
**7.Complex Attributes: -**

- For an entity, if an attribute is made using the multi valued attributes and composite attributes then it is known as complex attributes.

Ex: - A person can have more than one residence; each residence can have more than one phone number.

8.Key Attributes: -

- It represents a primary key.
- It is the unique characteristic of the entity.
- It is represented with 

**9.Non-Key Attributes: -**

- The attributes other than the primary key in a table are called non-key attributes.

10.Required Attributes: -

- A required attribute is an attribute that must have a value i.e., it cannot be left empty.

11.Optional Attributes: -

- An optional attribute is an attribute that does not require a value and it can be left empty.

4. Write about Relationship Degree?

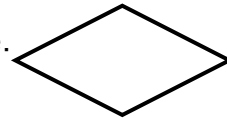
Relationship Degree:

Relationship: -

- It represents an association among several entities.

Relationship type: -

- The way in which two (or) more entities are related is called as relationship type.
- Relationship is represented with diamond shape.



Relationships based on degree: -

The number of participating entity types in a relationship is called Degree of Relationship.

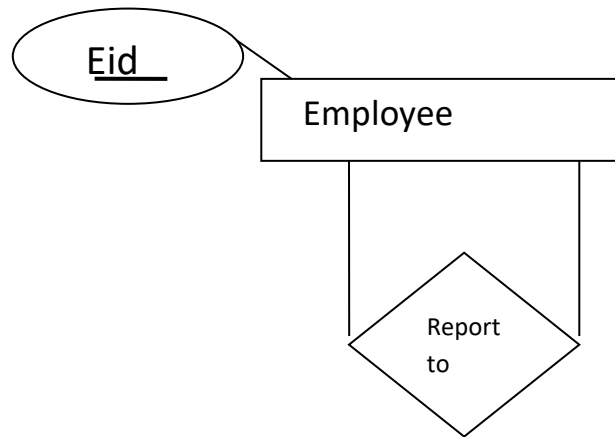
Types: -

The most common types of relationships based on degree are:

- Unary relationship
- Binary relationship
- Ternary relationship
- N-ary relationship

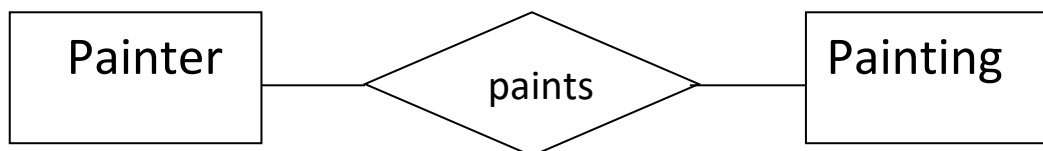
1. Unary relationship: -

- A relationship type of degree 1 is called unary relationship.
- When both the participants in the relationship belongs to the same entity, then the relationship is called as Unary relationship.



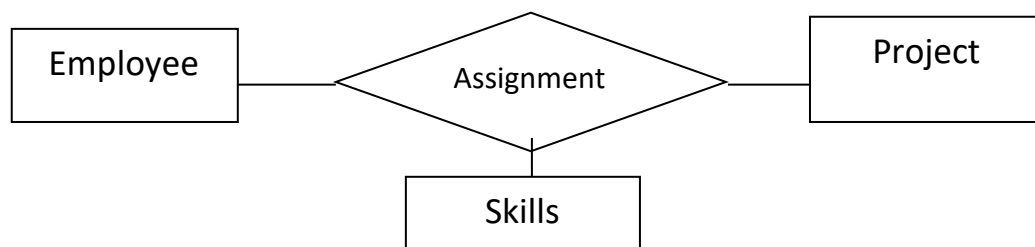
2.Binary relationship: -

- A relationship type of degree 2 is called binary relationship.
- When two entities participated in the relationship, then the relationship is called as Binary relationship.



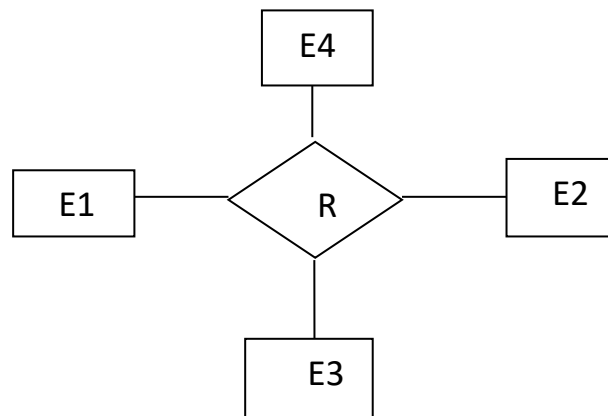
3.Ternary relationship: -

- A relationship type of degree 3 is called ternary relationship.
- When three entities participated in the relationship, then the relationship is called as Ternary relationship.



4.N-aryrelationship: -

- A relationship type of degree n is called as N-ary relationship.
- When 'N' number of entities participated in the relationship, then the relationship is called as N-ary relationship.



5. Explain about the classification of Relationship.

A. Relationship:

- It represents an association between several entities.
- A relationship works by matching data in key columns i.e., columns with same name in both the tables.
- The relationship matches the primary key in one table and foreign key in another table.

Types of Relationship:

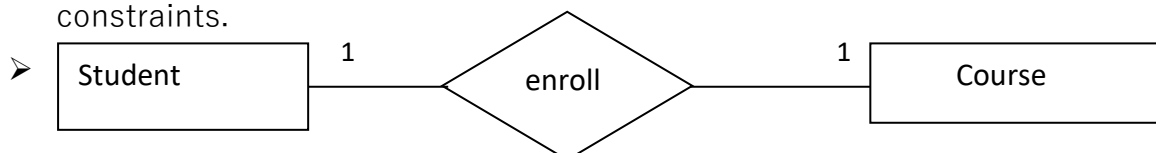
The type of relationship that is created depends on “How the related columns are defined.”

There are four types of relationships in the table. They are:

- One-to-one relationship (1..1)
- One-to-many relationship (1..m)
- Many-to-many relationship (m..m)
- Many-to-one relationship (m..1)

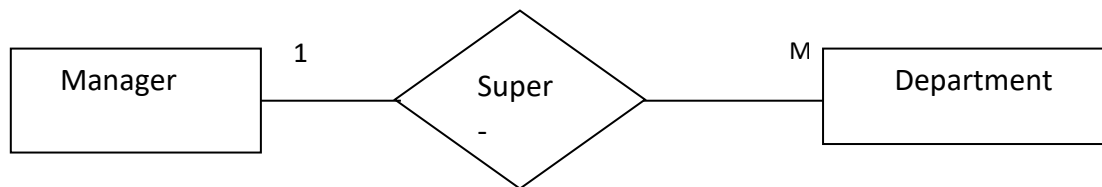
One-to-one relationship:

- In this relationship, a row in table ‘A’ can have only one matching row in table ‘B’ and vice-versa.
- It is created, if both of the related columns are primary keys or unique constraints.

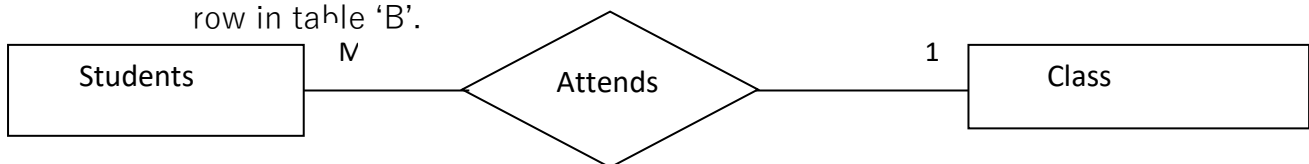


One-to-many relationship:

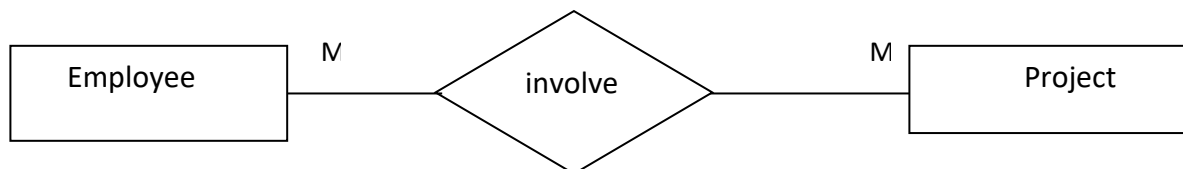
- In this relationship, a row in table 'A' can have many matching rows in table 'B', but a row in table 'B' can have only one matching row in table 'A'.

**Many-to-one relationship:**

- In this relationship, many rows in table 'A' can have only one matching row in table 'B'.

**Many-to-many relationship:**

- In this relationship, many number of rows in table 'A' can have many matching rows in table 'B' and vice-versa.

**6.Explain about EER model?****Enhanced/Extended Entity Relationship Model (EER) :**

Enhanced ER is a high-level data model that incorporates the extension to the original ER model. It is a diagrammatic technique for displaying the following concepts:

1. Sub-class and super class
2. Generalization and specialization
3. Union or category
4. Aggregation

Features of EER Model:

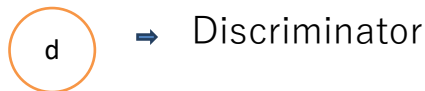
- It creates a design that can be more accurate to database schemas.
- It includes all modeling concepts of ER model.

- Diagrammatic technique helps for displaying EER schema.
- It includes the concepts of specialization and generalization.
- It is used to represent a collection of objects that is union of objects of different entities. Types:

1. **Sub-class and super class:**

Sub-class and super class relationship lead the concept of inheritance.

The relationship between sub-class and super class is denoted by

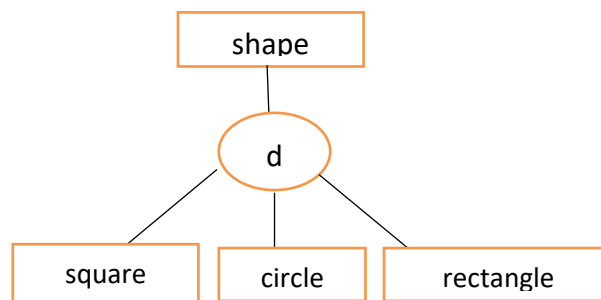


➤ **Super class:**

It is an entity type that has a relationship with one or more subtypes.

➤ **Sub-class:**

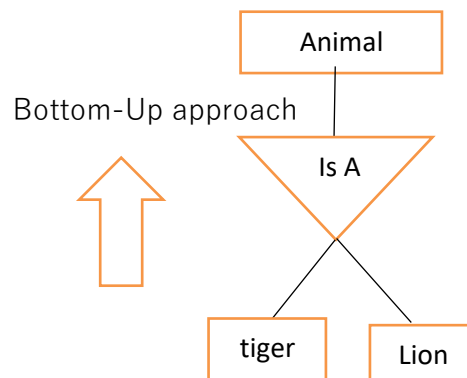
- It is a group of entities with unique attributes.
- It inherits properties and attributes from its super class.



2.Generalization and specialization:

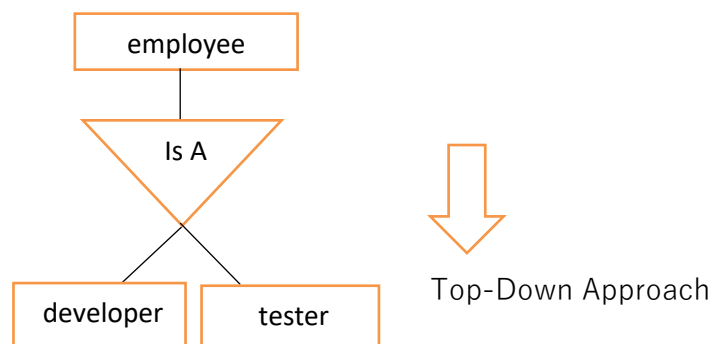
➤ **Generalization:**

- It is the process of extracting shared characteristics from two or more classes and combining them into generalized superclass.
- It is a bottom-up approach, in which two or more lower-level entities are combined to form a higher-level entity.
- It defines a general entity type from a set of specialized entity types.
- In this, super class is called as general entity and sub-classes are special entities.
- It minimizes the difference between entities by identifying the common attributes or features.



Specialization:

- It is a process that defines a group of entities which is divided into subgroups based on their characteristics.



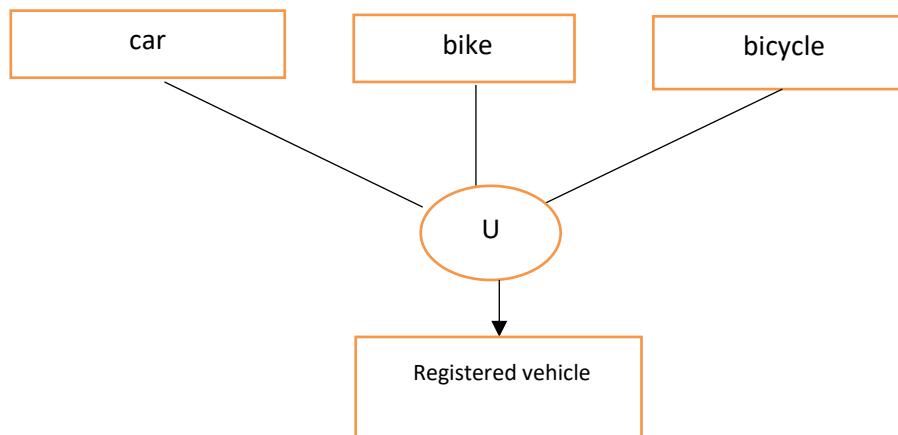
Here, Is A relationship specifying the inheritances.

- It is a top-down approach, in which one higher level entity can be broken down into two or more lower-level entities.
- It defines one or more sub-classes for the super class and also forms the super class/ sub-class relationship.
- It maximizes the difference between the entities by identifying the unique attributes or features.
- It is the reverse process of Generalization.

3.Union or Category:

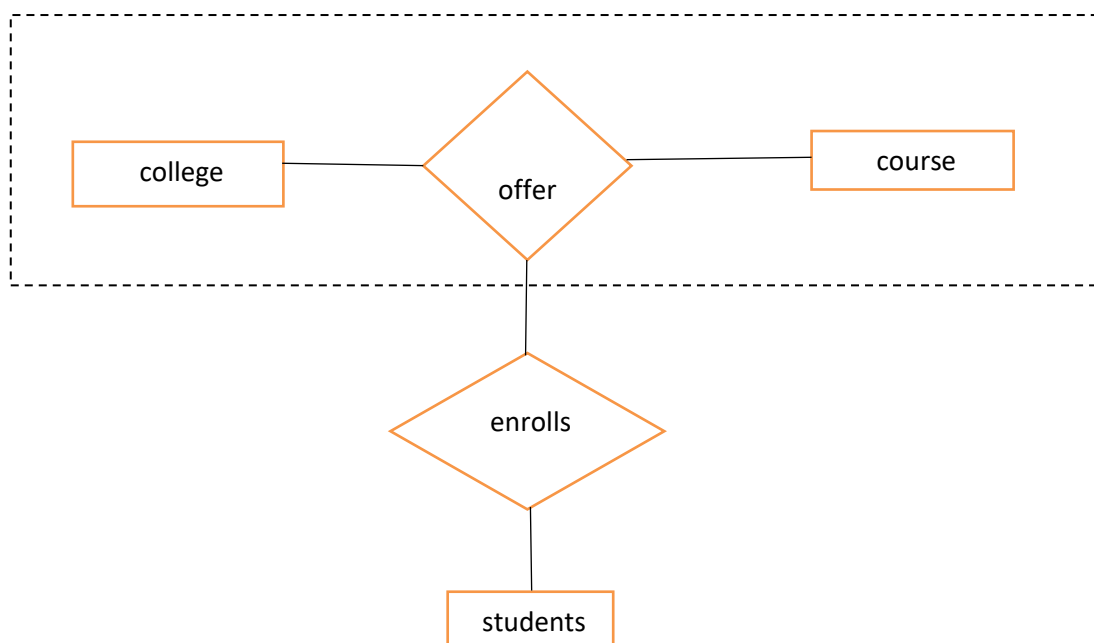
- It represents a single sub-class relationship with more than one super class.

- It can be a total or partial participation.
- A category member must exist at least one of its super classes.



4.Aggregation:

- It is a process that represents a relationship between a whole object and its component parts.
- When two entities are treated as a single entity in a relationship, then the relation is called as aggregation. It is used for data abstraction to hide the details of set of objects.



In the above example, the relation between college and course can act as a single entity which is in relation with student.

7.Explain about constraints on generalization and specialization?

A. Constraints on Generalization:

Generalization: It is the process of extracting the shared characteristics from two or more sub classes, then combining them into the generalized super class.

Specialization: It refers to creating new sub-classes from existing class with unique characteristics.

Constraints: The different types of constraints on generalization and specialization are:

1. Predicate defined or condition defined constraints
2. Attribute defined constraints
3. Disjoint and overlapping
4. Completeness constraints
5. Specialization hierarchies and lattice.

1.Predicate defined constraints:

- It refers to placing the condition on a value of attribute which identifies sub-classes.
- It specifies that, which members of sub-class can satisfy the predicate.

Ex: Secretary sub-class by defining the predicate job type= secretary

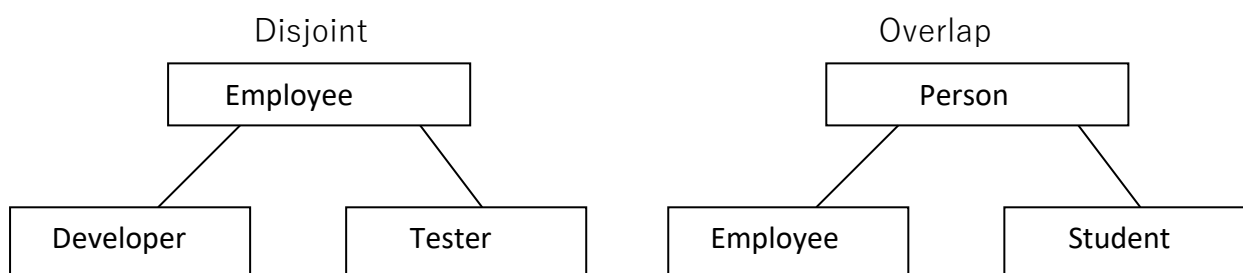
2.Attribute defined constraints:

- It specifies that, all sub-classes having membership condition on same attribute of super-class.

Ex: Job type

3.Disjoint and overlapping constraints:

- It specifies that sub-classes of specialization must be disjoint i.e., an entity belongs to at most one sub-class only.
- If sub-classes are not disjoint, then set of entities may overlap.



4.Completeness constraints:

- It specifies that, whether each entity super-type occurrence must also be a member of at least one sub-type.
- There are two types of completeness constraints
 - Total completeness
 - Partial completeness

Total completeness:

- It means that, every supertype occurrence is member of subtype.

Partial completeness:

- It means that not every super type occurrence is member of sub type.

5.Specialization hierarchies and lattice:

The sub-class itself can have further sub-classes specified on it, forming hierarchy or lattice of specialization.

A specialization hierarchy provides the means to:

- I. Support attribute inheritance
- II. Define disjoint/ overlapping and completeness/ partial constraints.

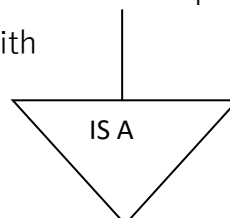
8.Write about Multiple inheritance in EER model?

Inheritance: -

- Inheritance is an important feature of generalization and specialization.
- Inheritance is represented with IS A relationship.
- IS A relationship is represented with

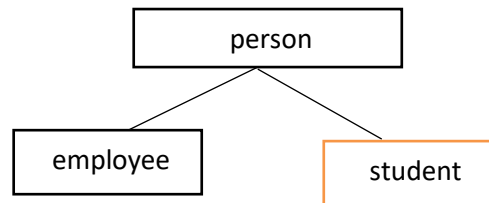
There are several forms of inheritance.

1. Single inheritance
2. Multiple inheritance
3. Repeated inheritance
4. Selective inheritance

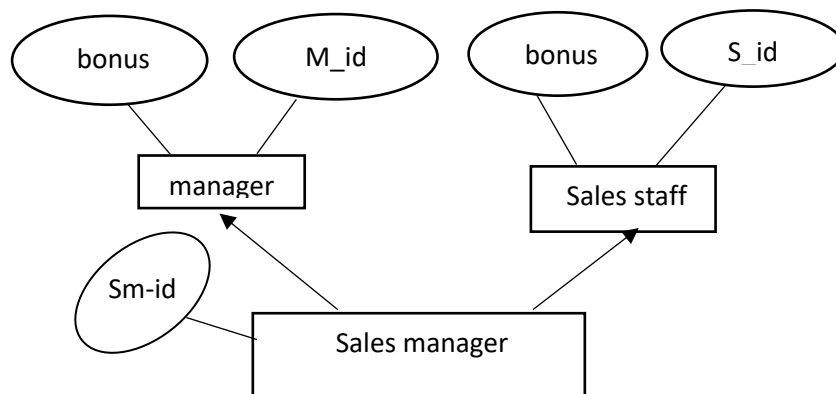


1.Single Inheritance: -

It refers that, the subclass inherits the attributes from only one superclass.

**2.Multiple Inheritance: -**

It refers that, the subclass inherits the attributes from more than one superclass.



The multiple inheritance is very problematic, the problem arises when the super classes contain the same attribute.

It can be handled in following ways.

1.It includes attributes and use the name of the superclass as qualifier.

For example, the super classes manager and the sales staff have the same attribute bonus then the subclass sales manager can inherit bonus has either **manager. bonus**(or)**sales staff. bonus**.

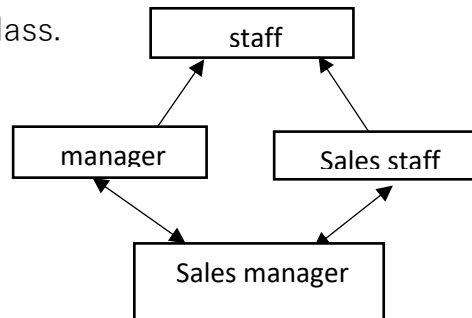
2.Use single inheritance to avoid the problem.

Ex: -sales manager → sale staff → manager
(or)

Sales manager → manager → sale staff

3.Repeated Inheritance: -

It is a special case of multiple inheritance in which the super classes inherit from a common superclass.



4.selective inheritance: -It allows the subclass to inherit a limited no. of attributes from subclass.

9.Write about Aggregation and Composition?

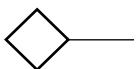
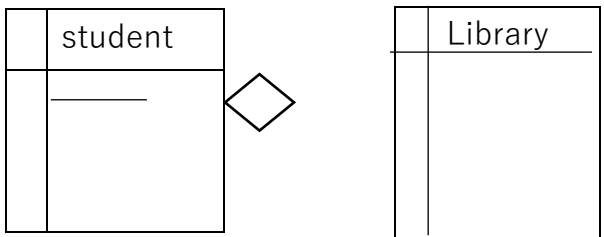
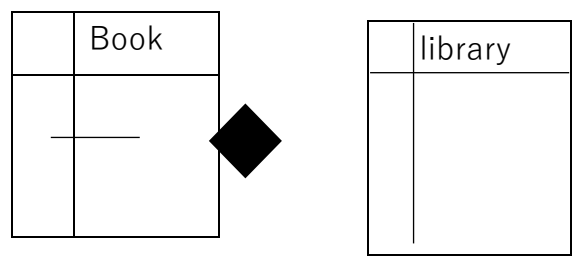
Aggregation: (Refer aggregation in EER model)

Composition: It is a special case of aggregation. A restricted aggregation is called as composition.

→ When an object contains another object, if the contained object cannot exist without the existence of container object, then it is called as composition.

Ex: If a class contains student, a student cannot exist without a class because there exists composition between class and student.

Difference between Aggregation and composition:

Aggregation	composition
1.It implies a relationship where the child can exist independently of the parent. 2.It defines a weak coupled relationship between two entities. 3.It is represented with  Ex: library(parent) Student (child). Delete the library and the student still exists. 4. When one entity could be part of another, but either can exist without the other independently.	1.It implies a relationship where the child cannot exist independently of the parent. 2.It defines a strong coupled relationship between two entities 3.It is represented with  Ex: library(parent) Book (child). Delete the library and the book doesnot exist. 4. When one entity could be part of another, but both need each other for their exists.
	

10. Write about advantages and disadvantages of ER modeling.**ER model:**

- It was introduced by Peterchen in 1976.
- It describes the structure of Database with the help of diagrams, which is known as ERD (Entity Relationship Diagrams).
- It is a blue-print or design of Database that can later be implemented as Database.

Advantages:

Following are the advantages of ER model.

1. Straight forward relation representation:

Having designed in ER diagram for database application, the relational representation of the database model becomes relatively straight forward.

2. Conceptually it is very simple:

Making the ER Diagram is very easy process.

3. Easy conversion from ER to other data models:

The conversion from ER diagram to any other data model can be easily accomplished.

4. Better Visual Representation:

An ER model gives graphical and diagrammatical representation of various entities, attributes and relationship between entities.

5. ER diagram constitute a very useful frame work for creating and manipulating database.

6. It is an effective communication tool.

7. ER diagrams are very easy to translate into relational tables which can be used to quickly build databases.

8. It is very efficient.

9. It is highly integrated with relational model.

10. Easy conversion to any data model.

Disadvantages:

Following are the disadvantages of ER model:

1. No industry standard for notation:

There is no industries standard notation for developing in ER diagram.

2. Popular for high level design:

The ER data model is especially popular for high level.

3. Only for relational data:

It is used to show the relationships. It shows only the relational structure.

4. Not for unstructured data:

It is only used for structured data.

5. It is difficult to exist with different architectures.

6. There are limited constraints and specifications in the ER diagram.

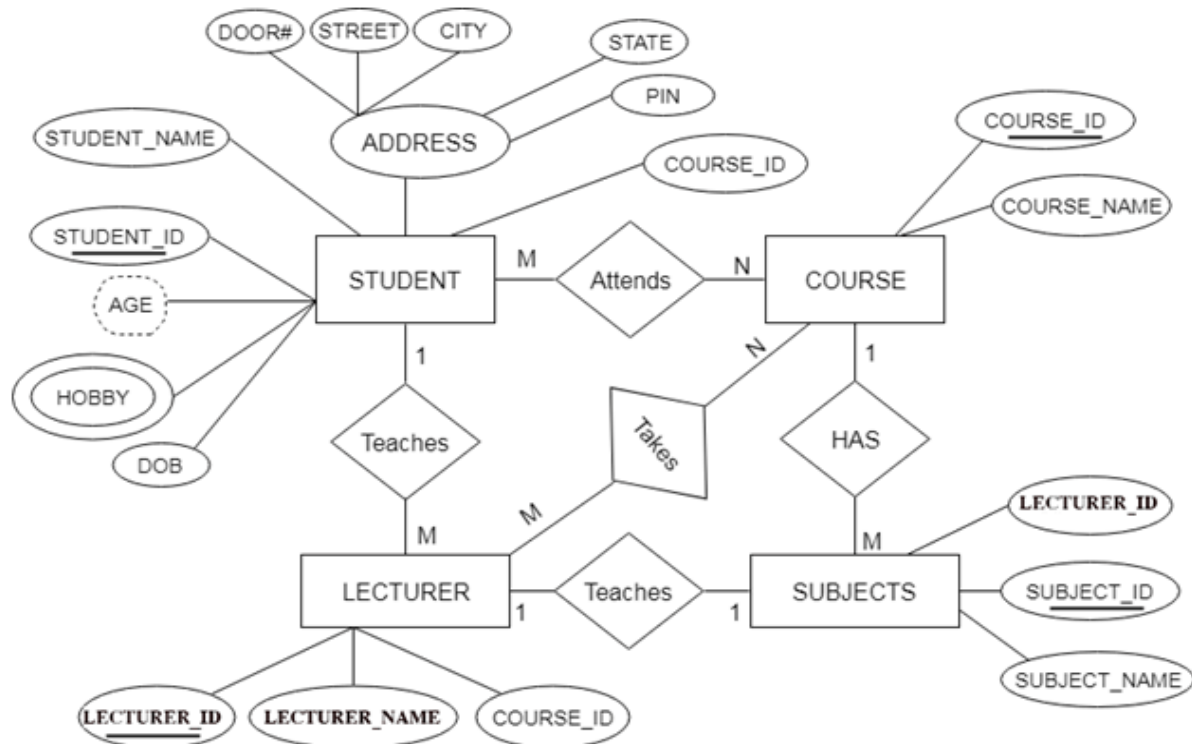
7. Limited relationships.

11. Explain about the conversion of ER diagrams to tables**A. Reducing ER diagrams to tables:**

ER Diagrams gives us the good knowledge about the requirement and the mapping of the entities in it. So, we can easily convert them as tables and columns.

- Using ER-Diagrams, one can easily create relational data model.

Ex: Consider an ER-Diagram of a college



- The basic rules for converting ER Diagrams into tables are:

Rule:1

- Convert all the entities in the diagram to tables.
- All the entities represented in the rectangle box in the ER-Diagram becomes independent tables in Database.

In the above diagram, student, course, lecturer, subjects form individual tables.

Rule:2

- All the single valued attributes of an entity are converted into a column of a table.

In the above diagram,
student (stu-id, sname)

course (course-id, course name)

lecturer (lect-id, lect-name)

subjects (sub-id, sub-name)

Rule:3

- All the key attributes in the ER-Diagram becomes the primary key of the tables.

In the above diagram, stud-id, course-id, lect-id, sub-id is primary key of respective tables.

Rule:4

- Declare the foreign key column if applicable.

In the above diagram, course-id and sub-id are foreign keys in their respective tables.

Rule:5

- Any multi-valued attributes are converted into new tables

In the above diagram,

- An attribute hobby is a multi-valued attribute.
- We cannot represent a multi-value in a single column in a student table.
- Hence, we create a separate table and stud-hobby with stud-id and hobby as its columns.
- We can create a composite primary key using both columns.

Rule:6

- Any composite attributes are merged into same table as different columns.

In the above diagram,

- Address is a composite attribute in a student table.
- Those attributes are merged into student table as individual columns.

Rule:7

- One can ignore derived attribute, since it can be calculated at any time.

Age can be derived at any point of time by calculating the difference between date of birth and current date. Hence, we need not create a column for this attribute.

→ It reduces the duplicity in database.

These are the very basic rules of converting ER Diagram into tables and columns and assigning the mapping between the table.

Rule:8

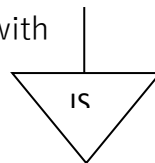
Converting weak entities:

- Weak entity is also represented as table.
- All the attributes of a weak entity form the column of the table.
- In our example, subjects are the weak entity.
- Hence, we create a table for it.
- Sub-id is represented as key attribute in the diagram, it cannot be considered as primary key.

12. Write about IS-A relationship and Attribute Inheritance

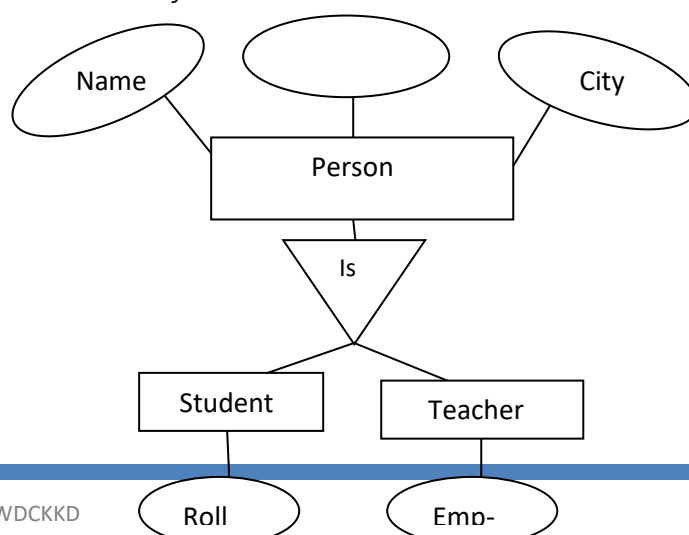
A. Inheritance:

- Inheritance is an important feature of generalization and specialization.
- It allows lower-level entities to inherit the attributes of higher-level attributes.
- Inheritance is represented with IS A relationship.
- IS A relationship is represented with



Attribute inheritance:

- A crucial property of the higher and lower-level entities which is created by specialization and generalization is attribute inheritance.
- The attributes of the higher-level entity sets are said to be inherited by the lower-level entity sets.



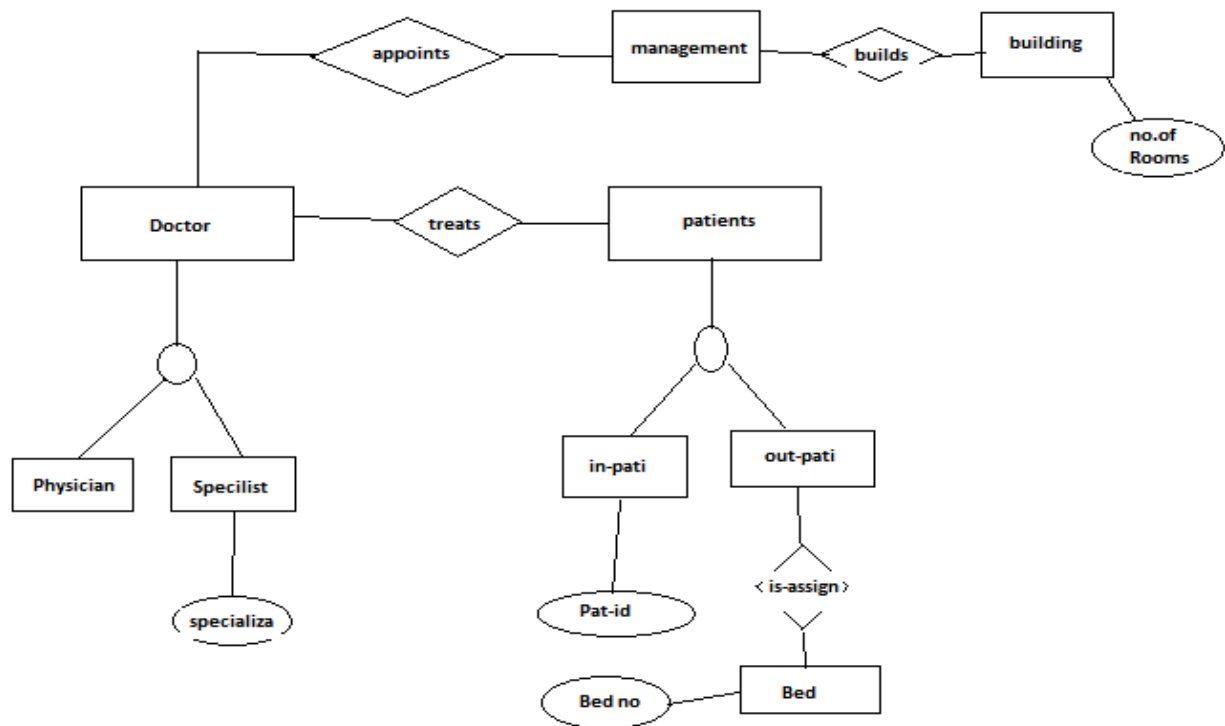
Here, the attributes of a person class such as name, gender, city can be inherited by lower-level entity such as student and teacher.

- Attribute inheritance applied through all tiers of lower-level entity sets.
- If the ER model was arrived at by specialization and generalization then the outcome is
 - A higher-level entity set with attributes and relationships that apply to all of its lower-level entity sets.
 - A lower-level entity sets with distinctive features that applied with only within a particular lower-level entity set.

13. Write about entity clustering.

Entity Clustering

- EER diagrams are difficult to read when there are many entities and relationships.
- One possible solution is to group entities and relationships into entity clusters.
- An entity cluster is a “virtual” entity type used to represent multiple entities and relationships in the ERD.
- An entity cluster is formed by combining multiple interrelated entities into a single abstract entity object.
- Entity cluster behaves like an entity type; hence entity clusters and entity types can be further grouped to form a higher-level entity cluster.



In the above EER model, management is responsible to appoint doctors of different categories, and also, they establish different buildings. Doctors treat patients who are classified into inpatient & outpatients. In the above diagram, there are so many entities. To reduce that complexity, we use entity clusters to doctor & patients.

